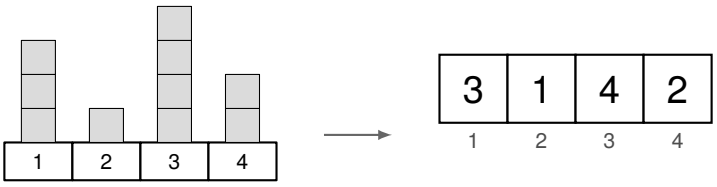
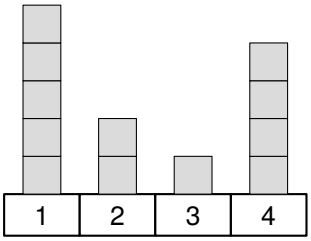
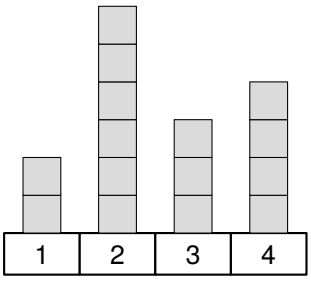
Build the towers on a strip of slots and play with a partner. Take turns. On your turn, take one whole tower from either end of the row. When the row is empty, the player with more cubes wins. The same number of cubes is a tie. Playing as well as you can means collecting as many cubes as you can be sure of. A row can be written with its heights:



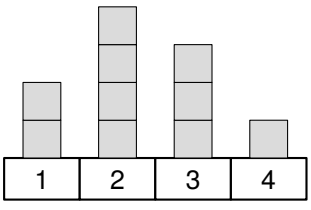
Problem 1: Play each row twice, so that each of you goes first once. Write both players' totals.



	went first	went second
game 1		
game 2		

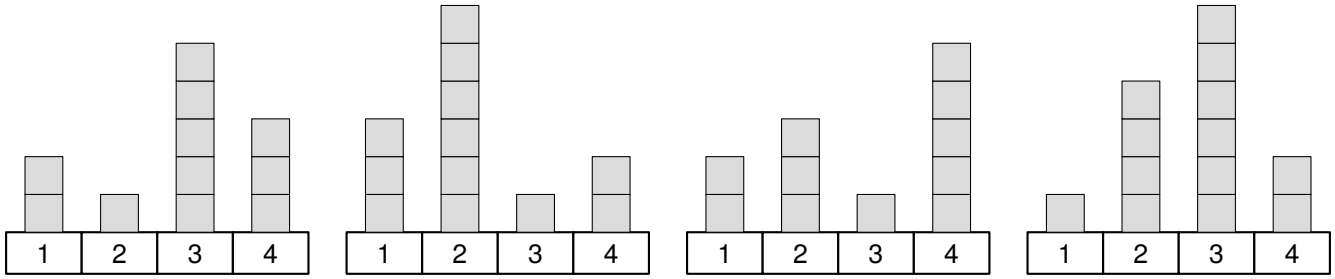


	went first	went second
game 1		
game 2		

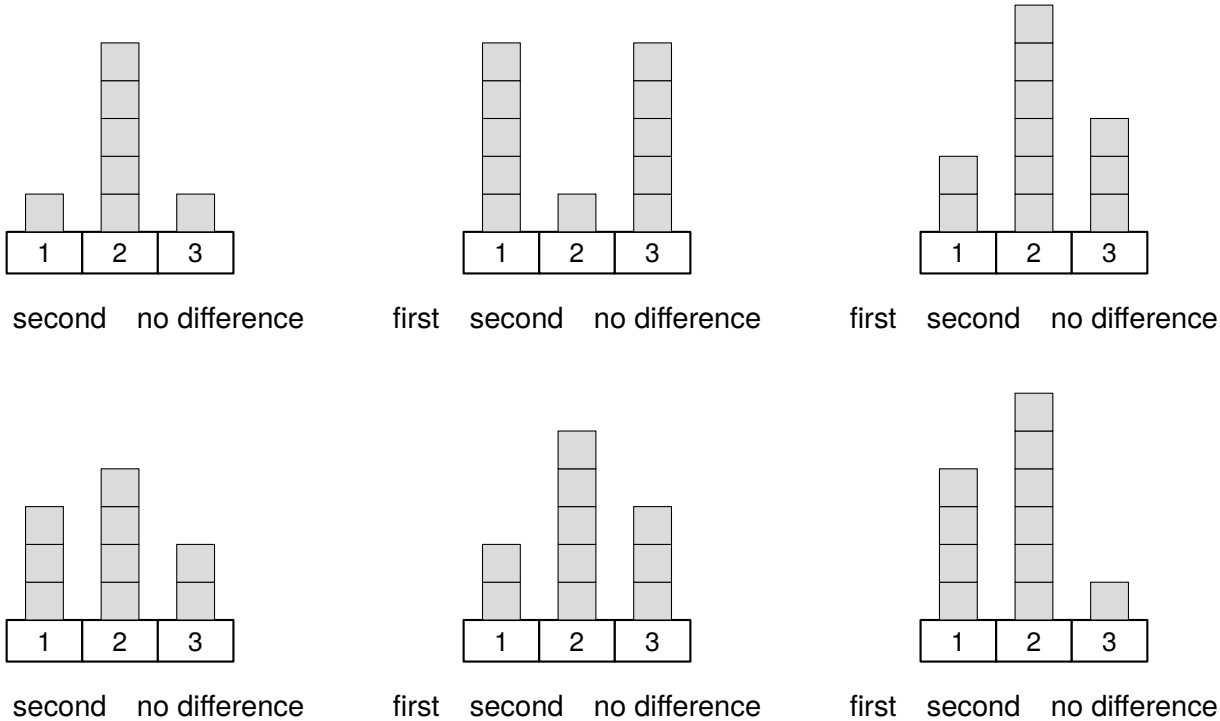


	went first	went second
game 1		
game 2		

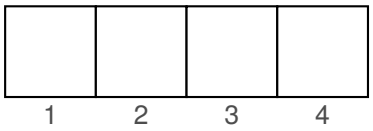
Problem 2: You go first on each of these rows. Find a first move that wins, however your partner plays after it. Ring that tower.

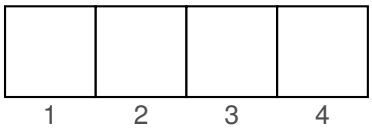


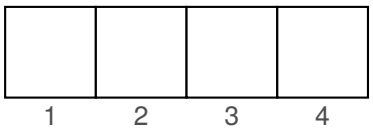
Problem 3: Is it better to go first or second on each of these rows? Ring your answer. Then find a way to tell from the three heights, without playing.

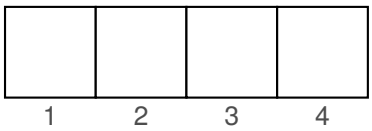


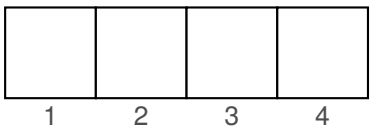
Problem 4: Can you find a row of four towers on which the first player loses, however the first player plays?

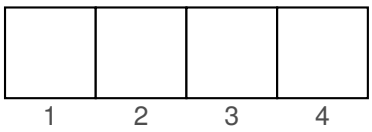


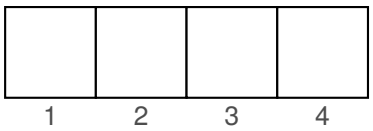


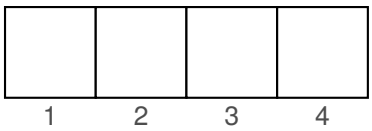




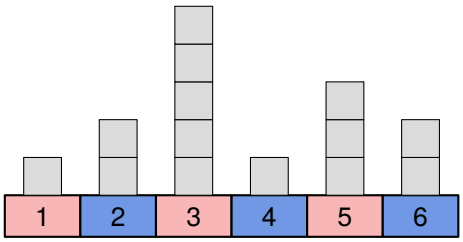


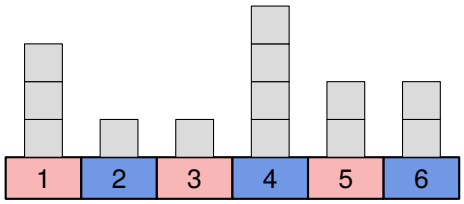


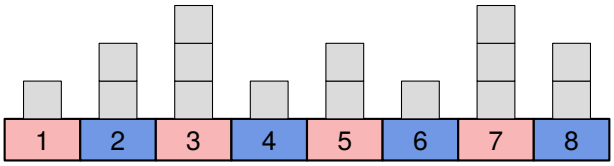




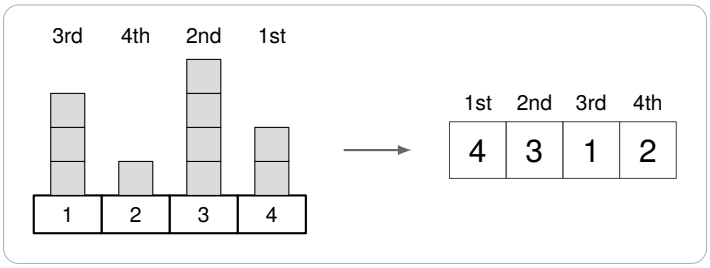
Problem 5: You and your partner play the grown-up together. The grown-up goes first on these colored rows. Can you beat the grown-up? How does the grown-up choose which towers to take?

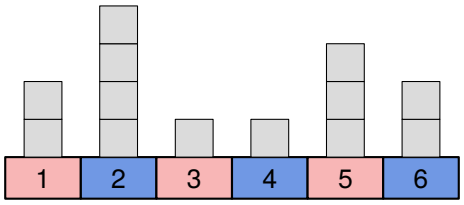


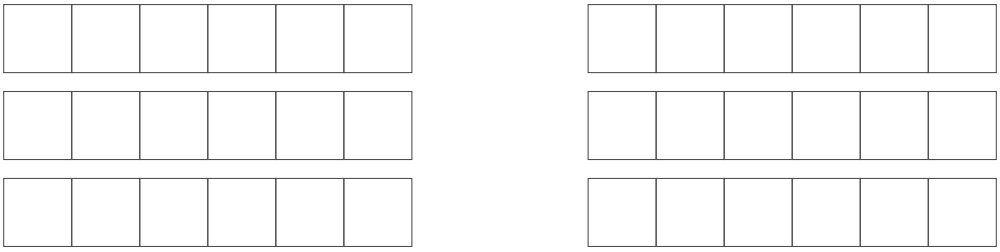




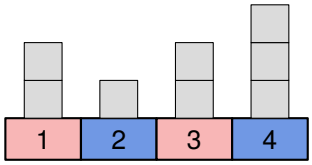
Problem 6: Go first on this colored row using the grown-up's plan. Can your partner ever make you take a tower on the other color? Try every way your partner can play, and write each game as the slot numbers in the order the towers were taken.



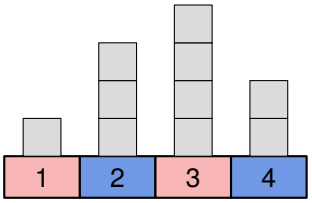




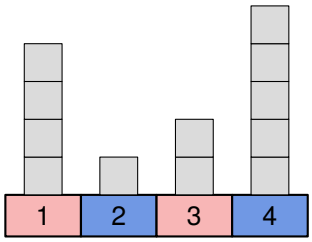
Problem 7: In each of these colored rows, the red slots and the blue slots hold the same number of cubes. Which rows can the first player win? Write the totals when both players play as well as they can.



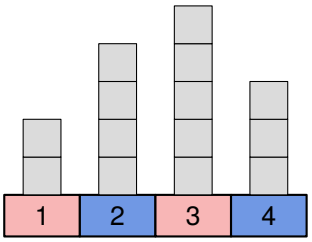
first _____ second _____



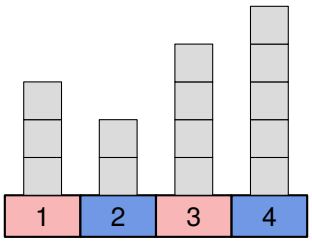
first _____ second _____



first _____ second _____

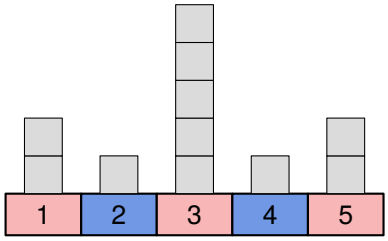


first _____ second _____

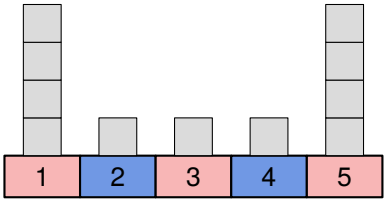


first _____ second _____

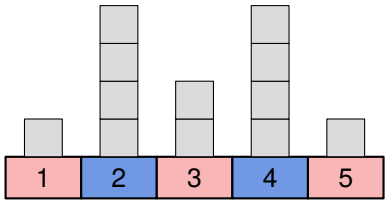
Problem 8: Does the grown-up's plan work on rows of five? Find out who wins each of these colored rows when both players play as well as they can.



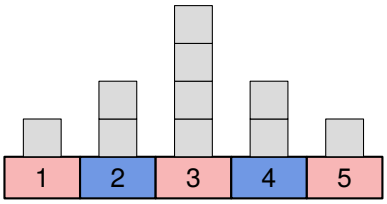
first _____ second _____



first _____ second _____



first _____ second _____



first _____ second _____

Problem 9: Make up a colored row of six where the red slots and the blue slots hold the same number of cubes, and the first player wins when both players play as well as they can.

